

Transport-Based Anatomical-Functional Metric Learning for Liver Tumor Recognition Using Dual-View Dynamic CEUS Imaging

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Abstract—Hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (ICC) are the most common malignancy arising from primary liver cancer (PLC). Liver ultrasound (US) has been the main approach for the early screening and differential diagnosis. Since ultrasonic findings of PLC are closely associated with tumor growth pattern, histological morphology, tumor staging, and other factors, ultrasonic imaging findings overlap partially. Besides, qualitative assessment is highly dependent on expertise. To improve the diagnostic objectiveness, we propose a novel transport-based anatomical-functional metric learning (T-AFML) method to *quantify imaging similarity* of both the gray-scale US and dynamic contrast-enhanced US view. Considering that the hemodynamic changes vary with individuals, we introduce a temporally regularized optimal transport to align the local enhancement patterns automatically. To sufficiently exploit ultrasonic findings similarity from different modalities, a selector-based metric integration mechanism is adopted to adaptively select a dominant modality accounting for the similarity measure. In this retrospective study, we collected a total of 174 liver cancer

patients consists of 105 HCC and 69 ICC, and our method achieves the superior diagnostic accuracy and sensitivity of 88.41% and 86.16%, respectively, demonstrating its efficacy in quantifying multi-modal ultrasonic findings similarity for PLC diagnosis.

Index Terms—Imaging similarity measure, optimal transport, primary liver cancer, temporal alignment.

I. INTRODUCTION

HEPATOCELLULAR carcinoma (HCC) and intrahepatic cholangiocarcinoma (ICC) are the most common malignancy arising from primary liver cancer (PLC), accounting for 75% and 12–15% of newly developed cases, respectively [1]. The early asymptomatic nature of these cancers frequently results in patients identified in the intermediate or advanced stages, leading to poor prognosis [2]. In addition, two types of liver cancer differ notably in surgical plans. Taking hepatectomy as example, a complete surgical resection of ICC generally suggests a larger resection margin than that of HCC, since that ICC is an aggressive cancer compared with HCC [3], [4]. Therefore, an accurate differential diagnosis of HCC and ICC is important to select treatment plan and improve outcome.

Liver ultrasonography has been the most widely used imaging tool for liver cancer screening. Various ultrasound (US) modalities exist with different clinical utility. Gray-scale US can sensitively detect focal liver lesions, differentiate benign or malignant based on morphologic findings, including tumor morphology, internal echo, boundary, and presence of acoustic halo at the edge, etc [5]. Previous studies reported that typical morphologic findings, such as the presence of peripheral bile duct dilatation and well-defined boundary [6], [7], could benefit liver cancer differentiation. Through peripheral intravenous injection of contrast agents, contrast-enhanced US enables to observe microvascular perfusion continuously and dynamically, as shown in Fig. 1, significantly improving the diagnostic resolution [8], [9], [10]. Enhancement characteristics (e.g., the starting time of enhancement, enhancement degree, and the spatial distribution of contrast agents, etc.) provide abundant information regarding tumor growth and progression [11]. Jin et al. [12], reported that the rate of peripheral hyperenhancement at arterial phase was significantly higher for ICC. Shin et al. [13], found a statistically significant difference of the exact timing of portal-phase

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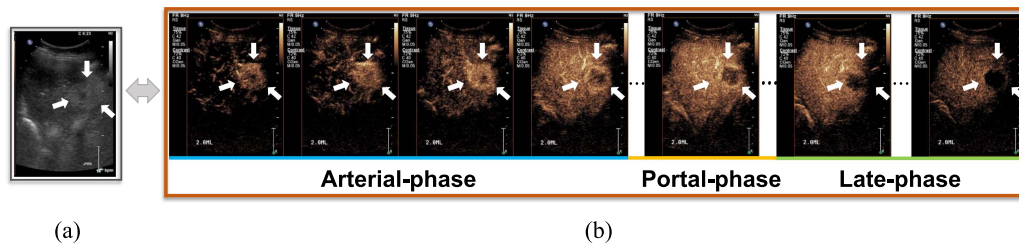


Fig. 1. An example of dual-view contrast-enhanced ultrasound (CEUS) imaging from an ICC case. (a) Gray-scale US image, which mainly characterizes the morphology information of lesion. (b) Contrast-enhanced US, which records the real-time blood supply within lesion across arterial-, portal-, and late-phase. Gray-scale US and Contrast-enhanced US are synchronous views, providing the complementary information of the same scanning plane.

wash-out between HCC and ICC, about 65 s vs. 35 s. However, either for gray-scale US or contrast-enhanced US, ultrasonic findings correlate with tumor growth pattern, histological morphology, tumor staging, and other complex variables [7], [9]. There exist overlapping imaging modes between HCC and ICC, for example, mass-forming ICC is regarded to have similar enhancement patterns with part of HCC since both involve the formation of mass [2], [14]. In addition, qualitative observation of ultrasonic features is subjective, resulting in inter-observer variability. Accordingly, there is an urgent need to develop an objective diagnostic method to improve the diagnostic efficacy of liver ultrasonography.

Recently, radiomics has achieved remarkable progress in clinics, especially in combination with machine learning techniques [15], [16], [17], [18], [19]. For clinical ultrasound images, radiomics attempts to establish the relationship between high-throughput features and liver lesions status. Kondo et al. [20] extracted 28-dimensional kinetic features from regional time-intensity curves (TICs), obtaining the accuracy of 87.7% for HCC and 85.7% for metastatic liver tumor. Further, manually-engineered imaging features [21], [22] or data-driven deep convolutional features [23] are exploited to characterize diverse enhancement patterns. Guo et al. [24] proposed a two-stage multi-view learning framework to fuse enhancement features from three vascular phases, differentiating the benign or malignant liver lesion. Zhang et al. [25] adopted the multi-source transfer learning framework to guide the knowledge transfer from contrast-enhanced US to gray-scale US. Among these studies, there is a lack of dynamic perfusion discrepancy measure and a joint similarity comparison of both gray-scale US and contrast-enhanced US.

In fact, clinical liver ultrasonography examination could be abstracted as patient similarity analysis [26]. Radiologists usually analyze ultrasonic manifestations from both gray-scale and contrast-enhanced US, compare them with diagnostic guidelines or clinical experiences, and finally make differential diagnosis after differentiation and reasoning. In terms of imaging analysis, radiologists identify the group of patients based on manifestation similarity of multiple ultrasound modalities. Motivated by this, we propose a metric learning-based method for the differential diagnosis of HCC and ICC. Specifically, our method quantifies the multi-modal ultrasound similarity of the target case with other patients with known pathological results, and diagnoses the liver cancer type as that has the maximum imaging similarity.

Essentially, it simulates the analogical reasoning process of senior radiologists during decision making.

Liver ultrasound-based patient similarity measure confronts with two major challenges. The primary challenge is *temporal alignment* of enhancement patterns when comparing pairs of dynamic CEUS sequences. Since the hemodynamic changes within liver are affected by multiple variables, including tumor angiogenesis, tissues necrosis, the degree of fibrosis, and fatty change, etc. [27], [28], [29], the velocity of blood flow always varies with different individuals [30], [31]. Obviously, a fixed point-to-point enhancement morphology comparison is hardly to capture local enhancement discrepancy. It is necessary to find the ideal inter-frame alignment relationship between local perfusion patterns, and take the weighted sum of local enhancement differences as the global similarity metric.

The second challenge is *metric fusion* of ultrasonic manifestation similarities from gray-scale US and contrast-enhanced US. Since the fact that typical imaging findings are not guaranteed to be observed in both ultrasound modalities simultaneously [7], we might confront the problem of inconsistent similarity measures between modalities. A possible example is that radiologists have observed distinct perfusion patterns in contrast-enhanced US, while could hardly identify the morphologic differences in gray-scale US. It is meaningful to design a clinically-inspired metric fusion strategy to tackle the inconsistency problem.

To tackle the abovementioned challenges, we propose a novel transport-based anatomical-functional¹ metric learning (T-AFML) method to quantify the ultrasonic similarity of gray-scale US and contrast-enhanced US for the differential diagnosis of HCC and ICC. The overall framework of this method is illustrated in Fig. 2. With the advantage of supervised metric learning, the linearly transformed features enable to capture important correlations of the raw radiomics features, resulting in the similarity measures consistent with pathological results. That is, the learned metric pulls the patients with the same type of liver cancer closer, while pushes the different types of patients far away, largely reducing the imaging overlapping of HCC and ICC. The main contributions are summarized as the two aspects.

- We introduce the temporally regularized optimal transport to resolve the temporal alignment of enhancement

¹For simplicity, we term the morphological and echoic characteristics in gray-scale US view as anatomical features, and dynamic enhancement patterns in contrast-enhanced US view as functional features.

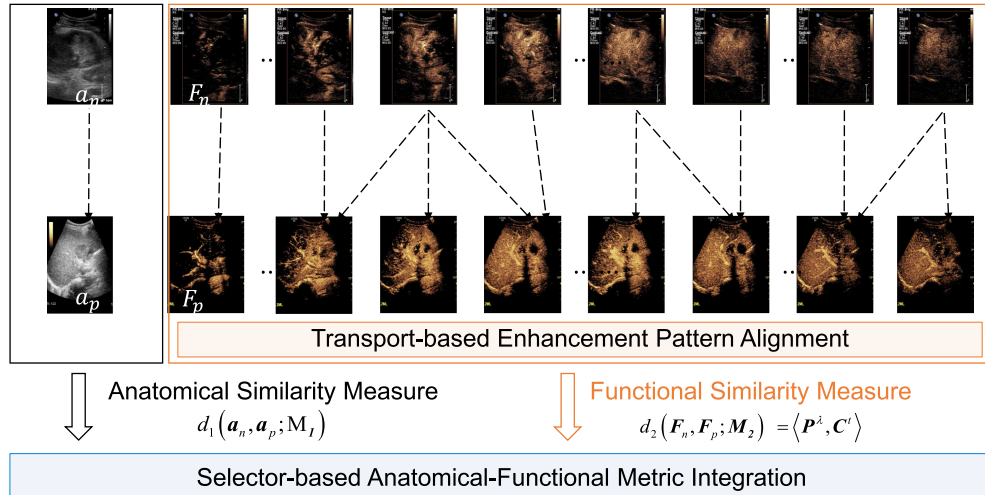


Fig. 2. An illustration of the proposed transport-based anatomical-functional metric learning (T-AFML) method. For the dual-view CEUS imaging, T-AFML enables to automatically align local enhancement patterns throughout the whole contrast-enhanced US sequences, upon which the global perfusion discrepancy $d_2(F_n, F_p; M_2)$ is measured. Then the quantified enhancement discrepancy is fused with the morphologic similarity $d_1(a_n, a_p; M_1)$ from gray-scale US to comprehensively characterize the ultrasonic findings similarity.

patterns. By casting the problem of perfusion sequence comparison into an optimal transportation problem, inter-frame alignment is derived on the enhancement morphology difference, time interval, and the respective diagnostic values.

- We introduce a selector-based fusion mechanism to integrate similarity metrics from gray-scale US and contrast-enhanced US. It is inspired by the clinical practice that radiologists actively select the more distinguished features to dominate the decision, for example, the presence of peripheral bile duct dilatation in gray-scale US highly suggests for an ICC case.

II. PRELIMINARY

A. Notations

Let $\{\mathbf{x}_n = [\mathbf{a}_n; \mathbf{F}_n], y_n\}_{n=1}^N$ be the training set containing N dual-view CEUS videos. Each sample $\mathbf{x}_n$ is composed of the morphologic representation $\mathbf{a}_n \in \mathbb{R}^{d_1}$ of the gray-scale US view, and a sequence of enhancement pattern representations $\mathbf{F}_n = [\mathbf{f}_1, \dots, \mathbf{f}_{I_n}] \in \mathbb{R}^{d_2 \times I_n}$ of the CEUS view. I_n is the length of input contrast-enhanced US sequences. $d_1(d_2)$ is the dimensionality of the respective view. y_n denotes the class label c_n (i.e., HCC and ICC). In the context of metric learning, we construct the supervised information in the form of triplets $\mathcal{R} = \{\mathbf{x}_n, \mathbf{x}_p, \mathbf{x}_q\}$, where the similar pair $(\mathbf{x}_n, \mathbf{x}_p)$ share the same pathological label and dissimilar pairs $(\mathbf{x}_n, \mathbf{x}_q)$ have the opposite diagnostic results. In our implementation, the number of similar/dissimilar samples for one patient is $k_p=3$ and $k_q=10$, respectively. For clarity, we have organized all notations in the Table I.

B. Optimal Transport

Optimal transport (OT) originates from the classic resource allocation problem [32], [33]. Suppose that an operator runs n

TABLE I
NOTATIONS

Variable	Description
$\mathbf{a}_n \in \mathbb{R}^{m_1}$	Morphologic representation from gray-scale US view
$\mathbf{F}_n = [\mathbf{f}_1, \dots, \mathbf{f}_{I_n}] \in \mathbb{R}^{m_2 \times I_n}$	The sequence of enhancement pattern representations from CEUS view
$\mathbf{x}_n = [\mathbf{a}_n; \mathbf{F}_n]$	Dual-view CEUS representation of the n_{th} sample
$\mathcal{R} = \{\mathbf{x}_n, \mathbf{x}_p, \mathbf{x}_q\}$	The triplet composed of the similar pair $(\mathbf{x}_n, \mathbf{x}_p)$ and dissimilar pairs $(\mathbf{x}_n, \mathbf{x}_q)$
T_{ij}	Time discrepancy of i_{th} and j_{th} enhancement pattern
C_{ij}	Enhancement morphology difference of i_{th} and j_{th} pattern
$\mathbf{C}^t = \mathbf{T} \circ \mathbf{C} \in \mathbb{R}_+^{I_n \times I_p}$	Spatial-temporal coupled metric
$\mu(\nu) \in \mathbb{R}^{I_n}(\mathbb{R}^{I_p})$	Marginal probability (i.e., weights) of dynamic enhancement pattern
$\mathbf{P} \in \mathbb{R}_+^{I_n \times I_p}$	Temporal alignment matrix between CEUS sequences
$d_1(d_2)$	Anatomical(Functional) similarity measure
$\mathbf{M}_1(\mathbf{M}_2) \in \mathbb{R}_+^{m_1 \times m_1}(\mathbb{R}_+^{m_2 \times m_2})$	Mahalanobis metric of anatomical(functional) similarity measure
$\mathbf{A}_1(\mathbf{A}_2) \in \mathbb{R}^{m_1 \times m_1}(\mathbb{R}^{m_2 \times m_2})$	The outer product of anatomical(functional) representation

warehouses and m factories, and each warehouse stores a certain amount of raw material needed by the factories. Assuming that the total amount of the material equals the total demand of the factories, the operator needs to figure out the most efficient transport plan to satisfy all demands of all factories. In this paper, we treat the dynamic enhancement discrepancy measure as an optimal transportation problem. Among which, the pair of liver perfusion sequences are viewed as all warehouses and all factories, and the distinguishing diagnostic values of local enhancement patterns are analogous to the amount of resources stored in warehouses (or demands from factories). Naturally, the transport cost is inter-frame enhancement discrepancy, consisting of the morphology differences (e.g., enhancement degree, texture, etc.) and the temporal interval between points. In

this way, we could compare dynamic enhancement discrepancy by computing a transportation plan from a source to a target contrast-enhanced US sequence. The weighted combination of the local enhancement difference is taken as the global perfusion metric, where weights is derived by OT.

Formally, suppose that we have two discrete distributions $\mu = \sum_i \mu_i \delta_{x_i}$ and $\nu = \sum_j \nu_j \delta_{y_j}$, and the given transport cost C_{ij} moving unit resource from x_i to y_j , the goal of optimal transport is to seek the most efficient transport plan to align μ to ν ,

$$d_{OT}(\mu, \nu) = \min_{P \in U(\mu, \nu)} \langle P, C \rangle \quad (1)$$

where $\langle P, C \rangle = \text{Tr}(P^T C)$ denotes the Frobenius dot-product of matrices P and C . $\text{Tr}(\cdot)$ outputs the trace of matrix. $U(\mu, \nu)$ is the set of all feasible plans, $U(\mu, \nu) := \{P \in \mathbb{R}_+^{n_1 \times n_2} | P \mathbf{1}_{n_2} = \mu, P^T \mathbf{1}_{n_1} = \nu\}$. Cuturi [34] introduced an entropic regularization to accelerate numerical computation. More importantly, regularized OT becomes strictly convex, solved by Sinkhorn-Knopp matrix scaling algorithm. Let $E(P) = \langle -P, \log P \rangle$ be the entropy of P , the regularized OT is formulated as

$$d_{reg}(\mu, \nu) = \min_{P \in U(\mu, \nu)} \langle P, C \rangle - \frac{1}{\lambda} E(P) \quad (2)$$

where $\lambda \geq 0$ balances the importance between the transport cost d_{OT} and the smoothness of P .

III. METHOD

A. Anatomical Similarity Measure

The goal of anatomical similarity measure is to induce a meaningful metric to quantify ultrasonic findings similarity in gray-scale US view. Under the supervision of pathological result, we expect the learned metric between patients diagnosed with the same liver cancer to be small as possible, while maximize the distance of patients with different diagnostic results [35]. Given radiomics-based representations ($\mathbf{a}_n, \mathbf{a}_p$) of morphology characteristics in gray-scale US, the squared anatomical similarity measure is defined as

$$d_1(\mathbf{a}_n, \mathbf{a}_p; \mathbf{M}_1) = (\mathbf{a}_n - \mathbf{a}_p)^T \mathbf{M}_1 (\mathbf{a}_n - \mathbf{a}_p) \quad (3)$$

where $\mathbf{M}_1$ is a positive semi-definite matrix defined by $\mathbf{M}_1 = \mathbf{L}_1 \mathbf{L}_1^T \in \mathbb{R}_+^{d_1 \times d_1}$. $\mathbf{L}_1$ is the linear projection of the raw anatomical feature space, which is expected to capture important correlations between multiple types of features. For simplicity, we define the outer product of data pairs $\mathbf{A}_1 = (\mathbf{a}_n - \mathbf{a}_p)(\mathbf{a}_n - \mathbf{a}_p)^T$, and re-parameterize the distance metric as $d_1 = \text{Tr}(\mathbf{M}_1 \mathbf{A}_1)$.

B. Functional Similarity Measure

1) Transport-based Enhancement Pattern Alignment:

Dynamic contrast-enhanced US allows radiologists to observe the progression of enhancement pattern continuously and dynamically, and focus on typical enhancement patterns regardless of the varying velocity of real-time blood flow. Therefore, temporal alignment is a prerequisite step to locate the pair of enhancement patterns for perfusion comparison, including new vessel formation, peripheral invasion, etc. According to

clinical experiences, an ideal temporal alignment should satisfy two conditions. *First*, the matched enhancement patterns are temporally adjacent, such that we compare the patterns from the same phase, e.g., the visibility of tumor boundary at the arterial phase, the degree of portal-phase wash out; It is obvious that arterial-phase enhancement degree should not be compared with that of portal-phase wash out (i.e., negative matching). *Second*, local enhancement patterns should be assigned with different weights, i.e., part of patterns deserve more chances to be compared to neighboring points. A simple intuition is that several typical patterns have higher diagnostic values and their differences should contribute more to the resulting metric. Note that, prior marginal probability allows for a unique advantage over DTW-like methods in our problem.

As mentioned in Section II-B, temporal alignment of enhancement patterns is casted as an optimal transportation problem. To encourage the matched patterns temporally neighboring, we add a temporal factor to rescale the enhancement morphology difference C , as follows,

$$C^t = T \circ C := [T_{ij} \cdot C_{ij}]_{ij} \in \mathbb{R}_+^{I_n \times I_p} \quad (4)$$

where T_{ij} is defined by the time interval of i -th and j -th pattern recorded in second,

$$T_{ij} = 1 + \beta |t_i - t_j| \quad (5)$$

where β adjusts the importance of temporal discrepancy in relative to the enhancement morphology difference. In this way, both enhancement morphology (i.e., enhancement degree, spatial distribution of contrast agents) and the onset of these patterns (i.e., the arrival time of contrast agents and the sustained enhancement time) could be jointly measured. β is set to 0.05 in our experiments.

Given pairs of dynamic enhancement representation $\mathbf{F}_n = [\mathbf{f}_1^n, \dots, \mathbf{f}_{I_n}^n]$ and $\mathbf{F}_p = [\mathbf{f}_1^p, \dots, \mathbf{f}_{I_p}^p]$, as well as the associated time point t_i^n (t_j^p), we could measure the local perfusion discrepancy between $\mathbf{f}_i^n$ and $\mathbf{f}_j^p$ as a weighted Mahalanobis distance,

$$C_{ij}^t = T_{ij} (\mathbf{f}_i^n - \mathbf{f}_j^p)^T \mathbf{M}_2 (\mathbf{f}_i^n - \mathbf{f}_j^p) \quad (6)$$

where C_{ij}^t is parameterized by $\mathbf{M}_2 = \mathbf{L}_2 \mathbf{L}_2^T \in \mathbb{R}_+^{d_2 \times d_2}$.

To take the distinguished diagnostic values into consideration, we assume that perfusion sequences $\mathbf{F}_n$ and $\mathbf{F}_p$ follow the discrete distributions μ and ν , where μ_i and ν_j denotes the respective importance weights, transport-based enhancement pattern alignment can be inferred by,

$$\begin{aligned} d_2(\mathbf{F}_n, \mathbf{F}_p; \mathbf{M}_2) &= \langle P^\lambda, C^t \rangle \\ &= \sum_{ij} P_{ij}^\lambda T_{ij} (\mathbf{f}_i^n - \mathbf{f}_j^p)^T \mathbf{M}_2 (\mathbf{f}_i^n - \mathbf{f}_j^p) \\ &= \text{Tr}(\mathbf{M}_2 \mathbf{A}_2) \end{aligned}$$

$$s.t. P^\lambda = \arg \min_{P \in U(\mu, \nu)} \langle P, C^t \rangle + \frac{1}{\lambda} \langle P, \log P \rangle \quad (7)$$

where $\mathbf{A}_2 = \sum_{ij} P_{ij}^\lambda T_{ij} (\mathbf{f}_i^n - \mathbf{f}_j^p)(\mathbf{f}_i^n - \mathbf{f}_j^p)^T$, P_{ij}^λ is the alignment weight assigned to the pair of pattern (i, j) . For the i -th pattern, the associated alignment weights satisfy the

marginal constraint $\sum_j P_{ij}^\lambda = \mu_i$. As seen from (7), a higher value μ_i mean that i_{th} enhancement pattern possesses more chance to be compared with other patterns, contributing more to the global perfusion metric. Motivated by the diagnostic practice that radiologists usually pay more attention to those points with relatively significant morphology changes, we propose to assign enhancement pattern weights according to their neighboring texture differences. With the sequence of inter-frame difference $[\Delta f_i^n, \dots, \Delta f_{I_n}^n]$, $\Delta f_i^n = f_i^n - f_{i-1}^n$, we obtain the normalized weights $\mu(\nu)$.

C. Anatomical-Functional Metric Integration

Suppose we have obtained the well-learned distance metric from the gray-scale US view and contrast-enhanced US view, the next challenge is how to integrate modality-specific metrics for an effective similarity measure. Considering the fact that different ultrasound modalities essentially characterize the target lesion from complementary views, similarity measures are supposed to be consistent with pathological results jointly. To this end, we adopt a selector-based metric integration method to select a dominant modality accounting for similarity measure. Given the learned anatomical metric d_1 and functional metric d_2 defined in (3) and (7), the selection of the dominant modality for the similar/dissimilar pairs is defined as

$$\begin{aligned} d_{AF}(\mathbf{x}_n, \mathbf{x}_p) &= \max(d_1(\mathbf{a}_n, \mathbf{a}_p), d_2(\mathbf{F}_n, \mathbf{F}_p)) \\ d_{AF}(\mathbf{x}_n, \mathbf{x}_q) &= \min(d_1(\mathbf{a}_n, \mathbf{a}_q), d_2(\mathbf{F}_n, \mathbf{F}_q)) \end{aligned} \quad (8)$$

In the training stage, for the patients diagnosed with the same liver cancer type $(\mathbf{x}_n, \mathbf{x}_p)$, the modality with larger distance is selected as the dominant modality by $\max(\cdot)$, which is expected to be shortened by parameter updates; while for the opposite pair $(\mathbf{x}_n, \mathbf{x}_q)$, the modality with smaller distance is selected by $\min(\cdot)$, which is supposed to be enlarged to cohere with pathological results. In this way, we expect to learn a consistent similarity metric across the gray-scale US and contrast-enhanced US. For our T-AFML method, patients are diagnosed as the same liver cancer type only when metrics of both ultrasound modalities are small as possible, and determined as the different types once any of views shows large discrepancy.

In the testing stage, given a testing sample $\mathbf{x}_j = [\mathbf{a}_j; \mathbf{F}_j]$, we first compute its $k = 3$ nearest neighbors based on each metric. With the smallest $k \times 2$ squared distance values, we label the testing sample by majority voting. In case of equal voting, the class with a smaller sum of distance values is used. This is motivated by the clinical practice that radiologists usually prefer the more typical features as the diagnostic basis, for example, the presence of peripheral bile duct dilatation in gray-scale US highly suggests for ICC even though enhancement features seem to be similar.

D. Objective Function

With the metric integration mechanism, a typical triplet loss of the dual-view CEUS sequences can be formulated as

$$J(\mathbf{M}_1, \mathbf{M}_2) = \frac{1}{|\mathcal{R}|} \sum_{n,p,q} \mathcal{L}(d_{AF}(\mathbf{x}_n, \mathbf{x}_q) - d_{AF}(\mathbf{x}_n, \mathbf{x}_p))$$

Algorithm 1: T-AFML.

Input: A training set of dual-view CEUS sequences $\{\mathbf{x}_n = [\mathbf{a}_n; \mathbf{F}_n], y_n\}_{n=1}^N$, and the associated triplets $\mathcal{R} = \{\mathbf{x}_n, \mathbf{x}_p, \mathbf{x}_q\}$.

Output: The Mahalanobis metric of gray-scale US and contrast-enhanced US, $\mathbf{M}_1$ and $\mathbf{M}_2$;

- 1: Initialize the Mahalanobis metric $\mathbf{M}_1, \mathbf{M}_2$, and temporal alignment matrix $\mathbf{P}^\lambda$ for all sequence pairs
 - 2: **while** J has not converged **do**
 - 3: **Learning the Mahalanobis Distance Metric**
 - 4: Compute the gradient of distance metric $\mathbf{M}_1$ and $\mathbf{M}_2$ by (11)
 - 5: Gradient update and semi-definite projection of $\mathbf{M}_1$ and $\mathbf{M}_2$ by (12)–(13)
 - 6: **Learning the Enhancement Pattern Alignment**
 - 7: Find the temporal alignment matrix $\mathbf{P}^\lambda$ defined in (14) for all pairs of CEUS sequences via Sinkhorn iteration (15)
 - 8: **end while**
-

$$+ \frac{\rho}{2} \sum_k h(\mathbf{M}_k) \quad (9)$$

where the objective function $J(\mathbf{M}_1, \mathbf{M}_2)$ is constituted by two terms: one is the triplet loss formed with a smooth hinge loss $\mathcal{L}(\cdot)$ that decreases monotonically, and the other is Frobenius norm $h(\mathbf{M}_k) = \|\mathbf{M}_k\|_F^2$ imposed to control the model complexity. ρ is the trade-off parameter between the two terms, which is set to 10^{-3} in our experiments. $|\mathcal{R}|$ denotes the number of all triplets, $k \in \{1, 2\}$ denotes the gray-scale US and contrast-enhanced US view, respectively.

Motivated by the work [36], [37], we adopt the smoothed hinge loss function $\mathcal{L}(\cdot)$ to facilitate distance learning for dual-view CEUS imaging, which is defined as

$$\mathcal{L}(x) = \begin{cases} \frac{1}{2} - x, & x < 0 \\ \frac{1}{2}(1 - x)^2, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases} \quad (10)$$

which separates differently labeled samples at a margin as possible. As reported in [38], a smoothed version of hinge loss is more effective for optimization.

E. Optimization

In this study, we adopt an alternating optimization strategy to minimize the whole objective in (9). More specifically, alternative updates can be divided into two steps as follows,

- Fix $\mathbf{P}^\lambda$ to update $\mathbf{M}_k$: learning the Mahalanobis distance metrics (ground metric).
- Fix $\mathbf{M}_k$ to update $\mathbf{P}^\lambda$: learning the enhancement pattern alignment.

1) Learning the Mahalanobis Distance Metric: When $\mathbf{P}^\lambda$ is fixed, the sub-problem turns into the Mahalanobis-like metric learning problem. For the l -th triplet, we use $k_l^+ \in \{0, 1\}$ to denote which modality is selected to measure the similarity of the similar pair. k_l^- is used for the dissimilar pair $(\mathbf{x}_n, \mathbf{x}_q)$ with

analogous meanings. The gradient of (9) w.r.t. $\mathbf{M}_k$ is

$$\frac{\partial J}{\partial \mathbf{M}_k} = \frac{1}{|\mathcal{R}|} \sum_{l \in \Gamma_l} \frac{\partial \mathcal{L}(k_l^- \text{Tr}(\mathbf{M}_l^- \mathbf{A}_l^-) - k_l^+ \text{Tr}(\mathbf{M}_l^+ \mathbf{A}_l^+))}{\partial \mathbf{M}_k} + \rho \mathbf{M}_k \quad (11)$$

where Γ_l is the subset of triplets whose element treats k -th metric as the determinant similarity measure either for the similar or dissimilar pair, *i.e.*, $\Gamma_l = \{l | k_l^+ = 1, \text{ or } k_l^- = 1\}$. $\mathbf{A}_l^+$ ($\mathbf{A}_l^-$) denotes the outer product of pairwise samples. Based on the derivative $\frac{\partial J}{\partial \mathbf{M}_k}$, we perform the gradient descent to update $\mathbf{M}_k$, and the t -th iteration is written as

$$\hat{\mathbf{M}}_k^{t+1} \leftarrow \mathbf{M}_k^t - \eta^t \frac{\partial J}{\partial \mathbf{M}_k^t} \quad (12)$$

where η^t denotes the step-size at the t -th iteration. We initialize the η^t to 1 and dynamically halve it when J does not decrease. We would not skip the current step of gradient update until the learning rate is lower than 0.005. After each gradient update, a projection step is performed on the current solution of (12) to obtain a well-defined Mahalanobis metric by singular value decomposition (SVD),

$$\mathbf{M}_k^{t+1} = \text{Proj}(\hat{\mathbf{M}}_k^{t+1}) = \mathbf{U} \max(\boldsymbol{\sigma}, 0) \mathbf{U}^T \quad (13)$$

where the positive eigen-values $\sigma_i \geq 0$ are preserved. $\mathbf{U}$ are the corresponding eigenvectors. In this way, we could maintain the positive semi-definite (PSD) property of Mahalanobis metrics.

2) Learning the Enhancement Pattern Alignment: When $\mathbf{M}_2$ is fixed, the sub-problem turns into the enhancement pattern alignment problem, $\mathbf{P}^\lambda$ is obtained by solving the following problem.

$$\mathbf{P}^\lambda = \arg \min_{\mathbf{P} \in U(\boldsymbol{\mu}, \boldsymbol{\nu})} \langle \mathbf{P}, \mathbf{C}_t(\mathbf{M}_2) \rangle + \frac{1}{\lambda} \langle \mathbf{P}, \log \mathbf{P} \rangle \quad (14)$$

Based on the Sinkhorn's theorem [34], the smoothed transportation plan has the following form $\mathbf{P}^\lambda = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v})$, where $\mathbf{K}_{ij} = e^{-\lambda \mathbf{C}_{ij}}$ is the element-wise exponential of $\lambda \mathbf{C}_{ij}$, $\mathbf{u}$, $\mathbf{v}$ are the scaling vectors which are solved by fixed-point iteration as follows

$$\begin{aligned} \mathbf{u}_{t+1} &\leftarrow \boldsymbol{\mu} / (\mathbf{K} \mathbf{v}_t) \\ \mathbf{v}_{t+1} &\leftarrow \boldsymbol{\nu} / (\mathbf{K}^T \mathbf{u}_t) \end{aligned} \quad (15)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the given enhancement pattern weights described in Section III-B. $\mathbf{u}$, $\mathbf{v}$ are initialized as uniform vector $\mathbf{u} = \frac{1}{I_n} \mathbf{1}_{I_n}$, $\mathbf{v} = \frac{1}{I_p} \mathbf{1}_{I_p}$. The iteration process will terminate when the change of the derived enhancement discrepancy does not exceed 0.01.

F. Implementation Details

Our method is implemented on MATLAB 2019b software running on a laptop with an Intel(R) Core i7 1.61 GHz CPU and 16 GB RAM. The running time is 145.82 s with the maximal number of alternate updates, and memory usage is 14 MB RAM. In our implementation, we solve enhancement pattern alignment $\mathbf{P}^\lambda$ first with an identity matrix to initialize Mahalanobis

TABLE II
DEMOGRAPHIC INFORMATION OF PATIENTS WITH HCC AND ICC

		HCC	ICC
Gender (F/M)		13/92	29/40
Age (year)	Median	62	63
	Mean	59.80±10.63	63.07±10.48
Size (cm)	Median	4.85	6.65
	Mean	5.69±3.44	6.95±3.41
	≤5	36	15

metric $\mathbf{M}_k$. The first step of pattern alignment is equal to find similar enhancement patterns in the original radiomic feature space. After obtaining the initial $\mathbf{P}^\lambda$, we continue this alternative optimization by (11)–(15). The maximal number of alternative updates is set to 10 and this process is terminated when the successive change of objective values is below 0.001.

IV. EXPERIMENTAL SETTINGS

A. Dataset

In this study, we totally collected 174 dual-view CEUS cine-loops from patients who attended Nanjing Drum Tower Hospital for liver ultrasound examination from January 2017 to October 2020. Among which, 105 cases with hepatocellular carcinoma (HCC) and 69 cases with intrahepatic cholangiocarcinoma (ICC) were included. Pathologies of all cases were determined based on biopsy or surgical specimens. Demographic information of patients was summarized in Table II. Each ultrasound radiologist involved in the acquisition of ultrasound images had more than 5 years of experience in abdominal ultrasound. A volume of 1.2 mL of the ultrasound contrast agent (SonoVue, Bracco, Milan, Italy) was injected into cubital vein in bolus, followed by a 5 mL saline flush. All examinations were performed on a Philips iU22 scanner (Philips Medical Systems, Best, the Netherlands) with a low mechanical index ($\text{MI} < 0.15$) to avoid the microbubbles disruption. The frequency of ultrasound scanner is 3.5 MHz. In clinical examination, the scanning acoustic window and depth were adjusted simultaneously to display the tumor and its surrounding liver parenchyma. To ensure adequate penetration, gain, dynamic range, mechanical index, output power, and focal zone and other parameters are jointly optimized.

For each sequence, enhancement process was recorded in seconds since the beginning of the saline flush, and lasted for at least 2 min until wash-out without changing the field of view. Apart from that, a synchronous B-mode US view corresponding to the same scanning plane was used to record the lesion location, size, shape, and echogenicity. The spatial resolution of each view is 600×400 pixels and the framerate is 9 fps. Approval was obtained by the ethics review board of local hospital and the informed consent was obtained from patients before this study.

B. Pre-Processing

1) Motion Compensation: Respiratory motion correction is one of preliminary pre-processing steps in the quantitative analysis of liver CEUS sequences, since target lesion regions

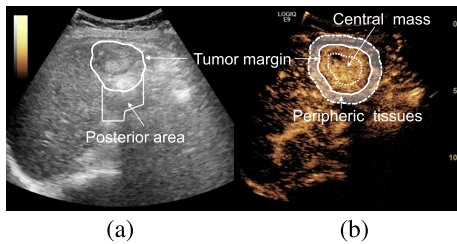


Fig. 3. An illustration of the partition of tumor regions both for gray-scale US and CEUS view. (a) Tumor region and the posterior acoustic region. (b) Tumor region, central mass, and the peripheral liver parenchyma.

always undergo irregular movements caused by free breathing, cardiac movements, and emotional fluctuation, etc. [39]. Inspired by the work [40], which reports that local feature descriptors could robustly characterize the “landmark” in ultrasound images, based on gray-scale US view, we adopt the compact and real-time descriptors (CARD) [41] as feature descriptors to estimate the affine transform for aligning successive frames. Besides, we extract the time-intensity curves (TICs) of the tumor regions after motion correction, and part of points are further removed when irregular fluctuations of TICs are detected.

2) Temporal Redundancy Elimination: For the most CEUS videos, sequential enhancement patterns appear to vary slowly in relative to the high temporal resolution. It is necessary to extract informative contrast frames without the information loss. Motivated by clinical quantitative analysis, we take the TIC as reference and divide the input sequence into two segments, *i.e.*, the wash-in phase and the wash-out phase (before/after perfusion peak). Among which, the starting point of wash-in phase and ending point of wash-out phase are determined by the point when the average pixel intensities reach 5% of the peak value of TIC. In addition to start-end points of each segment, we compute the inter-frame texture difference Δf_i within a small interval $2w + 1$ ($w = 1$) for each frame. Then, we sort them in the descending order and extract the top- k frames. Considering the faster variation in the former phase, we set the $k_{wash-in}$ to 8 and $k_{wash-out}$ to 5, respectively. Thus, a total of 16 frames are preserved for each CEUS sequence.

C. Imaging Feature Extraction

Based on the standardized guidelines defined in Ultrasound Liver Imaging Reporting and Data System (US LI-RADS) developed by the American College of Radiology [42], the major sonographic characteristics of a mass are grouped into morphology features, echogenicity features and posterior acoustic features. Towards a deep mining of each feature class, we also complement several widely used ultrasound imaging features, including tumor orientation and boundary features. Note that, echogenicity features are extracted both from tumor regions and their adjacent tissues since their differences are also of significance according to US LI-RADS (as shown in Fig. 3). In total, 145-dimensional anatomical features extracted from gray-scale US view and 152-dimensional functional features extracted from

contrast-enhanced US view. Detailed descriptions can be found in Supplementary file.

D. Evaluation Protocol

For the small-size dual-view liver CEUS dataset, we randomly split the whole dataset into two parts by preserving the class percentage, 80% for training and the remaining for testing. As random partition is involved, we repeat the evaluation procedure for 10 times and report the mean and standard derivation (std.) of testing results. As for model parameters selection, we perform a 4-fold cross-validation on the training data and determine the optimal parameter values that achieve the best mean accuracy on the validation sets. We fine-tuned model parameters in the first run and fixed them in the rest trials. In our experiments, ICC is labeled as the positive class since its lower diagnostic performance in clinics. The classification performance was evaluated by seven metrics, including Area under curve (AUC), classification accuracy (ACC), sensitivity (SEN), specificity (SPE), positive predictive value (PPV), negative predictive value (NPV), Matthews correlation coefficient (MCC), and F1-score. Note that, the prediction probability is estimated as the proportion of target category within nearest neighboring samples.

E. Competing Methods

1) Static Metric Learning Methods: We first compared our method with the static metric learning methods, *i.e.*, Information-theoretic metric learning (ITML) [43], Large margin nearest neighbor (LMNN) [44], and Unified multi-metric learning (UM²L) [45]. To integrate anatomical and functional characteristics, we adopted an ensemble mechanism (majority vote) to fuse distance metrics learned from dual-view CEUS imaging. Specifically, we extracted the point of perfusion peak as well as two points at which average perfusion intensity reaches 50% of the peak to cover three vascular phases.

2) Multi-view Learning Methods: T-AFML was further compared with several multi-view learning methods, including Deep canonical correlation analysis (DCCA) [46] and Autoencoder in autoencoder networks (Ae2-nets) [47]. As in [52], in the implementation of DCCA, the common intact representations learned from four US views are fed into a multiple-kernel SVM classifier for classification. As for Ae2-nets, classifier is implemented by a fully-connected layer with sigmoid activation. Similarly, we extracted the three typical frames as mentioned above and turns this into a four-view learning problem (1 B-mode US + 3 CEUS).

3) Temporal Fusion Methods: After that, we compared it with several mainstream temporal fusion methods, which are widely applied in video recognition and visual question analysis. 1) Non-local Neural Network (Non-local NN) [48]; 2) Multi-View Fusion Network (MVFNNet) [49] that adopts three independent 1D channel-wise convolutions over the temporal and spatial dimensions, conducting a spatio-temporal fusion. The proportion of input channels for multi-view fusion α is set to 0.25. As for Non-local NN, we treated the gray-scale US view as an additional point for dynamics fusion. While for MVFNNet, we added an independent branch for gray-scale US

TABLE III

CLASSIFICATION PERFORMANCES OF DIFFERENT COMPETING METHODS FOR THE DIAGNOSIS OF PRIMARY LIVER CANCER. THE TERMS A AND B IN “A ± B” DENOTE THE MEAN AND STANDARD DEVIATION

Method	Classification Metrics						
	AUC	ACC	SEN	SPE	PPV	NPV	F1-score
ITML [43]	76.37±2.53	73.68±2.56	61.45±5.22	81.75±4.27	69.21±4.52	76.36±2.13	64.89±3.70
LMNN [44]	80.58±4.42	76.09±5.06	66.30±2.92	82.54±8.55	72.76±8.98	78.68±2.49	69.03±4.24
UM ² L [45]	79.59±2.41	81.75±3.52	71.98±2.35	88.10±4.56	80.15±6.44	82.79±1.94	75.78±4.00
DCCA [46]	74.46±4.27	79.52±3.47	72.62±2.66	84.13±4.49	75.58±5.53	82.13±2.01	74.02±3.83
Ae2-Nets [47]	78.76±5.56	81.26±3.92	74.27±8.37	85.71±9.12	79.06±9.67	84.14±4.44	75.82±3.98
Non-local NN [48]	82.28±2.49	84.71±2.03	74.63±7.08	91.27±5.08	86.09±7.21	84.78±3.13	79.43±2.82
MVFNet [49]	84.49±3.34	85.98±1.17	80.13±8.35	89.68±5.08	84.17±5.37	87.99±4.04	81.55±2.53
SWMD [50]	76.38±4.88	80.38±2.59	66.30±9.67	89.68±3.27	81.29±3.99	80.47±4.07	72.45±5.61
RVSML [51]	86.90±1.63	87.62±2.69	83.33±5.32	90.48±2.75	85.47±3.66	89.17±3.20	84.29±3.56
T-AFML	87.91±2.25	88.41±1.43	86.16±6.52	90.06±2.24	87.76±2.26	89.81±3.29	85.23±2.61

view and performed the score-level fusion to combine modality information.

4) Temporal Alignment Methods: Finally, different sequence comparison methods were used for comparison. 1) The supervised word mover’s distance (SWMD) [50] learned OT distances for documents by minimizing the leave-one-out kNN error. In our experiments, the input of document is replaced by sequential contrast frames and initial weights are substituted by the prior weights described in Section III-B; 2) Regressive virtual sequence metric learning (RVSML) [51] adopts a similar OT variants to infer order preserving alignment between elements and extends the regressive virtual metric learning to the sequence setting. Similarly, gray-scale US view are viewed as an additional point for dynamics comparison.

Notably, we adopted the same pre-processing steps described in Section IV-B. Except for Ae2-Nets, Non-local NN and MVFNet, other competing methods are learned with the same radiomic features. Considering the limited size of our available data, initial weights of convolution blocks were learned by appending symmetric decoders, where each contrast frame was treated as an independent sample. For Ae2-Nets, Non-local NN and MVFNet, all US images are cropped by a bounding box that encloses the lesion boundary and resized to 224×224 pixels, among which, the median size of liver lesions is 157×168 pixels.

V. RESULTS

In this section, we first compare our T-AFML method with several state-of-the-art methods. Then, we validate the effectiveness of two important mechanisms of our method (i.e., transport-based enhancement pattern alignment and selector-based anatomical-functional metric integration), as well as the benefits of prior enhancement weights. After that, we further analyze the influence of the length of input contrast sequences and spatial-temporal metric parameters (Detailed content can be found in Supplementary file).

A. Comparison on Classification Performance

In this group of experiments, we evaluate the diagnostic performance of the proposed T-AFML for PLC (HCC vs. ICC) using dual-view CEUS sequences. To ensure a fair comparison,

both our method and competing method exploit ultrasonic characteristics from the static gray-scale US view and the dynamic CEUS view jointly. Quantitative classification results are presented in Table III.

Several issues can be summarized from Table III. First, the class of methods that perform dynamic enhancement evolution modeling (e.g., Non-local NN, RVSML, and T-AFML) generally outperform the other two classes of methods with relatively large margins, including the multi-view learning methods (e.g., Ae2-Net, DCCA) and static metric learning methods (e.g., ITML, LMNN). It demonstrates that the simplification of the complete perfusion similarity measure (enhancement modeling) into multi-view metric learning is insufficient to capture discriminative enhancement characteristics for PLC diagnosis. Although contrast frames are extracted in accordance with diagnostic practices (a single gray-scale US frame and three CEUS frames from the respective phases), the resulting diagnostic performance can only reach the mean F1-score of 75.0% by different levels of fusions, which is lower than the mean F1-score of 81.0% obtained by those methods taking the whole sequences as input, except for SWMD. The mean AUC of the former two classes of methods do not exceed 80.5%, while most methods performing dynamic enhancement modeling achieves AUC over 82.0%. This reveals that diagnostic-related enhancement characteristics not only consist in independent enhancement patterns but also in temporal dynamics, validating the importance of encoding the dynamic information of real-time blood supply in contrast-enhanced US view. As observed, the diagnostic performance of static metric learning methods is the weakest, even with the ensemble mechanism to fuse metrics leaned from different views. Except for UM²L, the mean accuracy and AUC of ITML and LMNN are 73.68% and 76.09%, 76.37% and 80.58%, respectively, and the sensitivity is only up to 66.3%. The lower performance can be explained by the totally ignored inter-frame dependency relationship in representation learning.

Second, we find that methods involving enhancement pattern alignment (e.g., RVSML and T-AFML) generally yield much better results among these temporal modeling methods (e.g., Non-local NN, MVFNet). The highest mean accuracy and F1-score are observed by the proposed T-AFML with 88.41% and 85.23%, respectively. Specifically, the performance improvements over RVSML with respect to AUC, ACC, SEN, PPV, MCC

TABLE IV

COMPARISON OF DIFFERENT TEMPORAL ALIGNMENT STRATEGIES ON THE TASK OF PRIMARY LIVER DIAGNOSIS. RESULTS ARE PRESENTED IN FORM OF [MEAN $\pm$ STANDARD DEVIATION]

Alignment	Method	AUC	ACC	SEN	SPE	PPV	NPV	MCC	F1-score
DTW	nDTW	80.84 $\pm$ 5.50	84.06 $\pm$ 1.39	71.52 $\pm$ 3.30	88.89 $\pm$ 4.49	81.59 $\pm$ 6.59	82.32 $\pm$ 1.71	62.08 $\pm$ 5.85	77.84 $\pm$ 1.55
	Soft-DTW	83.85 $\pm$ 3.71	86.12 $\pm$ 1.24	77.91 $\pm$ 6.69	88.89 $\pm$ 2.24	81.62 $\pm$ 3.36	86.15 $\pm$ 1.34	67.06 $\pm$ 3.18	81.39 $\pm$ 2.47
OT	OT	68.68 $\pm$ 2.51	72.28 $\pm$ 5.50	56.50 $\pm$ 11.41	85.71 $\pm$ 3.93	71.09 $\pm$ 5.34	74.36 $\pm$ 3.57	43.41 $\pm$ 2.49	61.58 $\pm$ 6.52
	Sinkhorn	76.18 $\pm$ 4.79	78.94 $\pm$ 1.52	62.55 $\pm$ 10.01	86.67 $\pm$ 1.90	76.00 $\pm$ 2.00	78.44 $\pm$ 1.16	52.38 $\pm$ 1.55	69.86 $\pm$ 3.92
	TL ^P	85.77 $\pm$ 3.61	86.60 $\pm$ 1.40	78.30 $\pm$ 4.17	89.12 $\pm$ 3.33	82.47 $\pm$ 4.61	86.85 $\pm$ 2.34	68.57 $\pm$ 4.19	82.22 $\pm$ 2.19
	OPW	86.98 $\pm$ 1.83	88.05 $\pm$ 1.89	84.34 $\pm$ 4.88	90.38$\pm$3.37	85.22 $\pm$ 4.17	89.44 $\pm$ 1.80	74.25 $\pm$ 4.46	84.87 $\pm$ 2.23
	T-AFML	87.91$\pm$2.25	88.41$\pm$1.43	86.16$\pm$6.52	90.06 $\pm$ 2.24	87.76$\pm$2.26	89.81$\pm$3.29	78.88$\pm$3.30	85.23$\pm$2.61

and F1-score are all statistically significant (p -values < 0.05). Meanwhile, we find the performance of SWMD deteriorates severely the F1-score of 72.45% even if access to the whole sequences, since it performed pattern alignment without considering temporal ordering. The mean AUC of temporal fusion methods (e.g., Non-local NN, MVFNet) can only reach 83.0%, while the temporal alignment methods can obtain around 85.0%. Experimental results show that pairwise patterns alignments based on spatiotemporal discrepancy is more effective to capture subtle enhancement differences in local temporal interval. In contrast, a global temporal aggregation into a vector-wise representation might lose significant enhancement differences, such as enhancement degree/homogeneity of in single pattern or a local window, or the changes of perfusion rate at different stages. Third, we find that there is a consistent gap between the sensitivity and specificity scores obtained by the competing methods and our T-AFML method. The highest mean sensitivity is observed for our T-AFML with 86.16% while the mean specificity score is 90.48% observed by RVSM. Since ICC is labeled as the positive class in our experiments, we could state that there is a higher possibility of misdiagnosing ICC cases as HCC. It is perhaps due to the reason that, as the most common malignant liver cancer, HCC often show diverse ultrasonic characteristics either in echogenicity of gray-scale US view or enhancement dynamics of CEUS view, leading to some degree of overlapping modes with those of ICC cases.

B. Comparison of Temporal Alignment Strategy

As introduced in Section III-B, a key component of our T-AFML method is the transport-based enhancement pattern alignment, which integrates the matching flexibility of optimal transport and temporal dependency of enhancement patterns. By incorporating the temporal discrepancy into the enhancement texture distance, T-AFML enables to compare perfusion differences both in spatial and temporal dimensions. To evaluate its effectiveness, we substitute this component by other temporal alignment strategies and compare their classification performances, including 1) DTW and its variant Soft-DTW that replaces the cost with a soft-minimum; 2) OT and its entropy-regularized version Sinkhorn; 3) two variants of OT with order preserving constraints (*i.e.*, TL^P [53] and OPW [54]).

The corresponding comparative results are presented in Table IV, from which we could have the following observations.

1) The OT and Sinkhorn based pattern alignments yield worse

diagnostic performances compared with other methods with temporal constraint. The performance degradation observed for OT is up to 19% in terms of accuracy. It is clear that the underlying temporal dependency of dynamic patterns is an indispensable diagnostic factor in contrast sequence comparison. In contrast, a totally flexible matching of enhancement patterns would result in biased similarity measures; 2) Three variants of OT (including our T-AFML method) generally outperform the other two variants of DTW by a margin of 3%. Among which, our T-AFML method and OPW achieve the best and second best classification results. The sensitivity and F1-score observed for OPW reach 84.34% and 84.87%, respectively. This might be owed to the higher matching flexibility of OT without the strict continuity and monotonicity constraint of DTW, since such constraints overemphasize the chronological order of patterns and ignore the possible differences of perfusion speed; 3) The proposed T-AFML method is still slightly superior to OPW on three metrics, which might be explained by the exact arrival time of each pattern rather than relative temporal position. In this way, we could take advantage of more precise temporal information into the ground metric.

C. Comparison of Metric Fusion Strategy

As introduced in Section III-C, in our multi-metric learning framework, *selector* was used to integrate anatomical and functional metrics, where $\max(\cdot)$ and $\min(\cdot)$ operator were used for similar and dissimilar pairs, respectively. In this group of experiments, we evaluated the effectiveness of this metric integration mechanism for dual-view CEUS sequence. Specifically, we conducted a two-fold evaluation, including 1) the comparison between the global weighted fusion (*i.e.*, mean or weighted fusion as introduced in Section III-C), and 2) the locally adaptive integration with the selector operators ($\min(\cdot)$, $\max(\cdot)$ operators for similar/dissimilar pairs).

Experimental results of different metric integration mechanisms are presented in Table V, from which we could observe that, the proposed T-AFML method with adaptive metric selection consistently lead to better diagnostic performances than the global modality fusion with fixed weights, and other three combinations of $\min(\cdot)$ ($\max(\cdot)$ operators). And the second best performance is achieved by the global weighted fusion with the learned weights 0.28/0.72 for the gray-scale US/CEUS view. The mean accuracy, sensitivity, and F1-score of the two methods are 87.52% vs. 88.41%, 78.21% vs. 86.16%, and 83.12% vs.

TABLE V
COMPARISON OF DIFFERENT FUSION BASELINES ON THE TASK OF PRIMARY LIVER CANCER DIAGNOSIS. RESULTS ARE PRESENTED IN FORM OF [MEAN $\pm$ STANDARD DEVIATION]

Fusion			AUC	ACC	SEN	SPE	PPV	NPV	MCC	F1-score
Global	mean		83.87±3.88	86.55±1.19	75.46±6.03	88.10±4.56	81.01±5.90	85.41±1.04	65.52±4.33	81.56±0.82
	weighted sum		85.30±1.61	87.52±2.03	78.21±5.39	90.48±2.05	83.97±1.64	86.36±3.10	69.27±0.73	83.12±2.90
Adaptive	Positive	Negative								
	min (·)	min (·)	84.73±4.13	85.64±0.16	73.53±5.11	89.29±3.95	82.02±5.76	85.28±1.16	66.04±3.46	80.20±1.09
	max (·)	max (·)	84.72±3.69	86.46±2.23	72.80±6.88	89.52±3.56	82.54±5.26	85.50±1.12	66.83±3.48	80.61±3.78
	max (·)	min (·)	80.50±5.66	83.81±1.35	69.05±6.73	88.10±4.12	79.94±6.38	82.20±1.66	60.52±5.74	77.23±2.27
	min (·)	max (·)	87.91±2.25	88.41±1.43	86.16±6.52	90.06±2.24	87.76±2.26	89.81±3.29	78.88±3.30	85.23±2.61

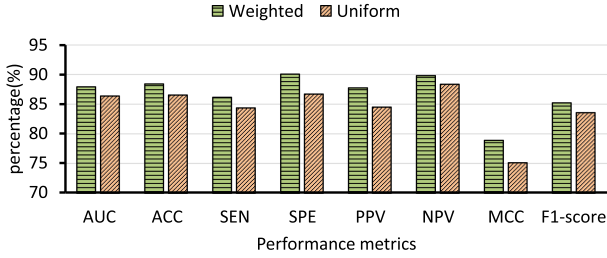


Fig. 4. Comparison between our T-AFML models trained with marginal constraints of prior weights and uniform discrete distribution.

85.23%, respectively. Nevertheless, except for our configuration of $\max(\cdot)(\min(\cdot))$ for similar (dissimilar) pairs, performance improvements are not observed for the other three types of selectors in comparison to a global fusion. Especially for the opposite combination ($\max(\cdot)(\min(\cdot))$ for similar (dissimilar) pairs), performance declines are up to 3.1% in terms of mean accuracy compared with the mean fusion. Experimental results validates that applying $\max(\cdot)/\min(\cdot)$ operators for metric selection of similar/dissimilar pairs is effective to sufficiently exploit metric information learned from complementary US modalities; From the point of optimization, selection mechanism enables the proposed T-AFML method to receive the useful gradient information from the corresponding pairs, thus guiding each Mahalanobis metric $\mathbf{M}_k$ to update toward the right direction.

D. Effectiveness of Frame Weights With Prior Knowledge

As introduced in Section III-B, we utilize inter-frame difference to determine the marginal probability distribution of contrast sequences in accordance with diagnostic practices. When no prior knowledge is available, marginal probability distribution is usually defined as a uniform discrete distribution $\mu = \frac{1}{I_n} \mathbf{1}_{I_n}$, $\nu = \frac{1}{I_p} \mathbf{1}_{I_p}$, assigning the same diagnostic value to all enhancement patterns. To verify the effectiveness of distinguished diagnostic values, we conducted another group of experiment with uniform marginal distributions, and compare the resulting performance with that of our T-AFML method.

The corresponding results are presented in Fig. 4, based on which, we can find that non-uniform pattern weights could obtain further performance gain in comparison with uniform distribution constraint that treats each pattern equally. Sensitivity and F1-score are increased from 84.36% to 86.16% and 83.58%

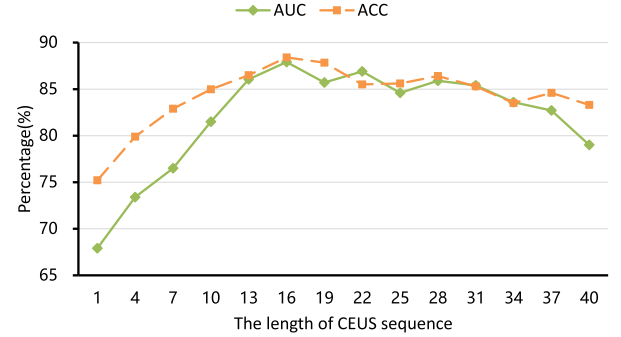


Fig. 5. Results of PLC classification obtained by our T-AFML method by changing the length of input contrast sequences L_c from 1 to 40.

to 85.23%, respectively. This is intuitive and reasonable, since the evolution rate of perfusion patterns are not uniform and part of patterns may contain significant texture variations within relatively short intervals. Thus, non-uniform marginal distribution would allow patterns with higher weights to outweigh other points, which is also in accordance with clinical diagnostic practices.

E. Influence of the Length of Dual-View CEUS Imaging

In this group of experiments, we investigated the influence of the length of input contrast sequences (denoted as L_c) on the perfusion discrepancy measure achieved by our T-AFML method. In the experiments, we orderly selected L_c from the range of [1–40] with a gap of 3. Among which, for the case $L_c = 1$, we selected the point to perfusion peak. For the other cases $L_c \geq 4$, we chose contrast frames as described in Section IV-B. Experimental results are quantified by the accuracy and presented in Fig. 5.

From Fig. 5, we can observe that the value of accuracy clearly increases when changing L_c from 1 to 16. The lowest accuracy is observed for the single peak frame with 75.21%, and the optimal performance is obtained by accepting contrast sequences with the length of 16 as inputs. This implies that too few enhancement patterns are not enough to yield satisfactory results, potentially because limited number of enhancement patterns cannot comprehensively characterize the global enhancement differences. Meanwhile, the performance of our method tends to be stable around 0.86 between $L_c = 16$ and $L_c = 28$, and gradually declines to 83.32% until L_c is up to 40. The similar

changing tendency can also be observed with the AUC curve in Fig. 5. Experimental results motivate us to choose $L_c = 16$ in our implementations, with the goal to largely include potentially informative patterns, as well as to account for the computational complexity during model training.

VI. DISCUSSION

A. Advantages Over Other Alignment Strategies

Learning similarity metric for dynamic medical imaging becomes much more challenging due to the inherent temporal dependency. Even for the similar imaging findings, temporal alignment is indispensable to associate points with the related semantics since the evolution speed or sampling rate is often different. Up to now, dynamic time warping (DTW) as well as its variants [55], [56], [57] has been applied most widely, which finds the alignment path via dynamic programming. Although dynamic programming-based methods are efficient to tackle the problem of the varying evolution speeds (i.e., local accelerations and decelerations), eliminating the difference of enhancement evolution rates completely might lose discriminant perfusion characteristics, since the real-time blood flow rates are associated with complex tumor histological patterns. In fact, a much earlier and marked wash-out of ICC is generally treated as a distinguishing characteristic from HCC [13]. In addition, these methods usually assume different time points possess the same importance, and the alignment path follows the temporal ordering and boundary matching strictly. These assumptions and constraints seem to be unnecessary for dynamic perfusion analysis. In the task of dynamic perfusion comparison, a reasonable and flexible alignment should be inferred on the enhancement morphology difference, time interval, and the respective diagnostic values.

B. Limitation and Future Work

In this study, we propose a metric learning framework for dual-view CEUS imaging, in which patient similarities are jointly measured by lesion anatomical and functional features. Although the proposed dual-view CEUS metric learning method achieved competitive performances in primary liver cancer diagnosis, there still exist some limitations to be solved to extend the clinical application and improve model performance.

First, the performance of our T-AFML method is only validated on the binary classification task (HCC vs. ICC), which is limited by the available pathological labels at hand. In fact, our method provides a metric learning framework for dual-view CEUS sequences, where differential diagnosis is implemented by nearest neighboring classifier. A natural extension is to perform multi-class classification by collecting liver cancer cases with different differentiation degrees, such as poorly differentiated, moderately differentiated, and well-differentiated stages. Second, our method lacks an effective measure of pattern alignment accuracy but the eventual triplet metric constraints. Considering the dramatically increased annotation burden of each pair of CEUS sequences, our method could be extended to a semi-supervised version in which part of sequence pairs

are labeled for alignment loss calculation. Third, in the current implementations, both anatomical and functional features from dual-view CEUS are represented by predefined handcrafted features independent of the subsequent metric learning process, which might lead to sub-optimal performance. It should be a promising direction to further unify feature extraction and metric learning into a deep learning framework.

VII. CONCLUSION

In this study, a transport-based anatomical-functional metric learning (T-AFML) method is proposed to measure ultrasonic differences of dual-view CEUS imaging. In order to quantify dynamic perfusion discrepancy, we introduced a transport-based enhancement pattern alignment strategy to compare perfusion discrepancy both in spatial and temporal dimensions. On the collected dual-view CEUS dataset with 174 subjects diagnosed as HCC or ICC, the effectiveness of our proposed method on joint anatomical and functional US similarity learning has been extensively evaluated. Compared with several state-of-the-art CAD methods and existing radiomic studies, the proposed method has demonstrated better or at least comparable classification performance.

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